FACULTY OF ENGINEERING TECHNOLOGY

Subject: co[redacted]smission — Mid Exam Spring 24/25

Name: [redacted] — Duration: 50 min

[redacted] EXAM CONTAINS TWO QUESTIONS

Q1:(Please write your answers in a table only, otherwise, your answers will be neglected)

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	d	c	B	B	D	A	B	B	A
Question	11	12	13	14						
Answer	A	D	A	C						

MECHATRONICS SPECTRUM

1. Error Control and Flow Control are the responsibilities of
   a) data link and network Layers b) Data Link and Transport Layer
   c) presentation Layer d) data link and Physical Layer

2. The SNR for ideal channel is:
   a) 0 b)0.5 c)1 d)∞

3. Queuing time is function of:
   a) load b) distance c) transmission d) phase

4. The bandwidth of digital signals used in data communications is:
   a) discrete with finite rate
   b) discrete with an infinite range
   c) continuous with finite range
   d) continuous with an infinite range

5. We send a voice signal from a microphone to a recorder, the transmission is:
   a) Baseband transmission
   b) Broadband transmission
   c) only voice, no band
   d) both a and b

6. The term that describes the position of the waveform relative to time 0 is
a) Wavelength b) Period c) Frequency d) Phase

7. The transmission impairment that refers to a signal with high energy in a very short time is:
a) Induced Noise b) Crosstalk c) Impulse Noise d) response noise

8. A signal with 200 milliwatts power passes through 10 devices each with an average noise of 2 microwatts. Calculate SNR:
a) 5000 b) 10000 c) 150 d) 250

9. Route determination can be identified by:
a) Transport layer b) Application layer c) Network layer d) Physical Layer

10. A signal in which 1 bit lasts 0.001 s, the Bit rate would be:
a) 1kbps b) 500bps c) 1700bps d) 1Mbps

11. National Internet Service Provider (ISP) networks are connected to one another by private switching stations called:
a) Network Access Points
b) Peering Points
c) National ISP
d) direct ISP

12. OSI stands for:
a) open system internet
b) Online Systems Interconnection
c) Online systems Interconnection
d) Open Systems Interconnection

13. Keyboard and traditional monitors are examples of:
a) Simplex devices
b) Duplex devices
c) Half Duplex devices
d) Full Duplex system

14. Two computers connected by an Ethernet hub are of:
a) TAN topology
b) MAN topology
c) WAN topology
d) LAN topology

Q2: A nonperiodic composite signal has a bandwidth of 240 kHz, with a middle frequency of 140 kHz and peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal

Bandwidth = High$_f$ − Low$_f$

240 = 380 − 140

Hz = 240

f_m = 240 = 260 − 20

High$_f$ = 260 K Hz

Low$_f$ = 20 K Hz



